HeartSense AI: An AI-Powered Heart Disease Prediction

Project Report
Submitted for the subject of
221601907 Capstone Project – II

Integrated Master of Science (IT) (IMSc (IT))

Semester IX

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Faculty of Computer Applications & IT
IMSc (IT) Programme
Ahmedabad

Academic Year 2026–2027

GLS UNIVERSITY
Faculty of Computer Applications & IT
IMSc (IT) Programme
Ahmedabad

CERTIFICATE

This is to certify that the following students of **Semester IX Integrated Master of Science (IT) (IMSc (IT)) Programme**, **Faculty of Computer Applications & IT (FCAIT), GLS University, Ahmedabad**, have successfully completed the subject project titled "**HeartSense AI: An AI-Powered Heart Disease Prediction"** for the course **221601907 Capstone Project – II - System Design and Implementation** during the **Academic Year 2026–2027.**

Date of Submission: 30-10-2026

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1. Brief of System

1.1. Introduction

Heart disease is one of the major health concerns worldwide. Early identification of potential risk factors and timely medical consultation can help improve patient care. Traditional assessment methods often require healthcare professionals to examine patient medical records, health parameters, and clinical test results before evaluating the possibility of heart disease.

HeartSense AI is a machine learning-based web application designed to estimate the likelihood of heart disease using patient health data. The system uses medical parameters such as age, sex, chest pain type, resting blood pressure, cholesterol level, resting electrocardiogram (ECG) results, maximum heart rate, and other relevant clinical features to generate a prediction.

The project uses machine learning algorithms such as Logistic Regression, Random Forest, Support Vector Machine (SVM), and XGBoost to train and compare predictive models. Based on the selected model, the system displays prediction results and, where supported, provides feature-importance explanations using Explainable AI techniques such as SHAP or LIME.

The application also includes user authentication, role-based access for patients, doctors, and administrators, patient data management, risk assessment, health education recommendations, interactive dashboards, and PDF report generation. These features help organize patient information and present model outputs in an understandable format.

The main objective of HeartSense AI is to develop an accessible, data-driven decision-support tool that assists users and healthcare professionals in understanding potential heart disease risk. The system is intended for educational and research purposes and does not replace professional medical advice, clinical diagnosis, or treatment.

1.2. Problem Stasement

Heart disease is one of the major health challenges worldwide, and identifying potential risk at an early stage is important for timely medical evaluation and preventive care. Traditional heart disease assessment relies on clinical examinations, medical history, and diagnostic tests, which require professional expertise to interpret.

The main problem addressed by HeartSense AI is the need for an accessible, data-driven system that can analyze relevant patient health parameters and estimate the likelihood of heart disease. Manual interpretation of multiple medical attributes can be time-consuming, and patients may find it difficult to understand the factors associated with their health risks.

The proposed system uses machine learning algorithms such as Logistic Regression, Random Forest, Support Vector Machine (SVM), and XGBoost to analyze patient data and compare prediction performance. It also aims to provide understandable explanations of predictions, categorize estimated risk levels, and present general preventive healthcare recommendations.

Therefore, the project focuses on developing a web-based heart disease prediction and clinical decision-support system with secure patient data management, explainable predictions, interactive dashboards, and PDF report generation. The system is intended to support educational and research activities and assist clinical review; it does not replace professional diagnosis or medical decision-making.

1.3. Objective of the System

The main objectives of HeartSense AI: An AI-Powered Heart Disease Prediction and Clinical Decision Support System are as follows:

1. Patient Data Collection: To allow users to enter relevant medical information, including age, sex, chest pain type, blood pressure, cholesterol level, ECG results, and maximum heart rate.

2. Secure User Authentication: To provide secure registration and login facilities with role-based access for patients, doctors, and administrators.

3. Patient Data Management: To enable authorized users to add, update, view, and manage patient medical records and maintain patient history securely.

4. Data Preprocessing: To handle missing values, remove duplicate records, encode categorical features, normalize numerical data, and prepare the dataset for machine learning.

5. Machine Learning Model Training: To train and evaluate machine learning algorithms such as Logistic Regression, Random Forest, Support Vector Machine (SVM), and XGBoost for heart disease prediction.

6. Model Performance Comparison: To compare the performance of different algorithms using evaluation metrics such as accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices.

7. Heart Disease Prediction: To analyze patient health parameters and estimate the likelihood of heart disease using the selected machine learning model.

8. Explainable AI (XAI): To explain model predictions using techniques such as SHAP or LIME and identify the health features that contribute to individual predictions.

9. Risk Assessment: To categorize prediction results into Low Risk, Moderate Risk, and High Risk using documented thresholds and display the associated prediction probability where supported.

10. Personalized Recommendations: To provide general health education, lifestyle suggestions, dietary guidance, exercise recommendations, and advice to seek professional medical consultation when appropriate.

11. Dashboard and Visualization: To display prediction results, patient statistics, medical parameter charts, prediction history, and machine learning model performance metrics through an interactive dashboard.

12. PDF Report Generation: To generate downloadable and printable reports containing patient details, prediction results, risk estimates, relevant explanations, and recommendations.

13. Prediction History Management: To maintain historical prediction records so that authorized users can review previous results and observe changes over time.

14. Administrative Management: To provide administrative facilities for managing user accounts, roles, system records, and application activities according to assigned permissions.

15. Data Security and Reliability: To protect sensitive medical information, enforce access controls, maintain data integrity, and ensure that predictions are generated from the correct patient inputs.

16. Clinical Decision Support: To develop an accessible, data-driven system that supports the understanding of potential heart disease risk while emphasizing that predictions are estimates and do not replace professional diagnosis or treatment.

1.4. Hardware and Software Requirements

1.4.1. Hardware Requirements

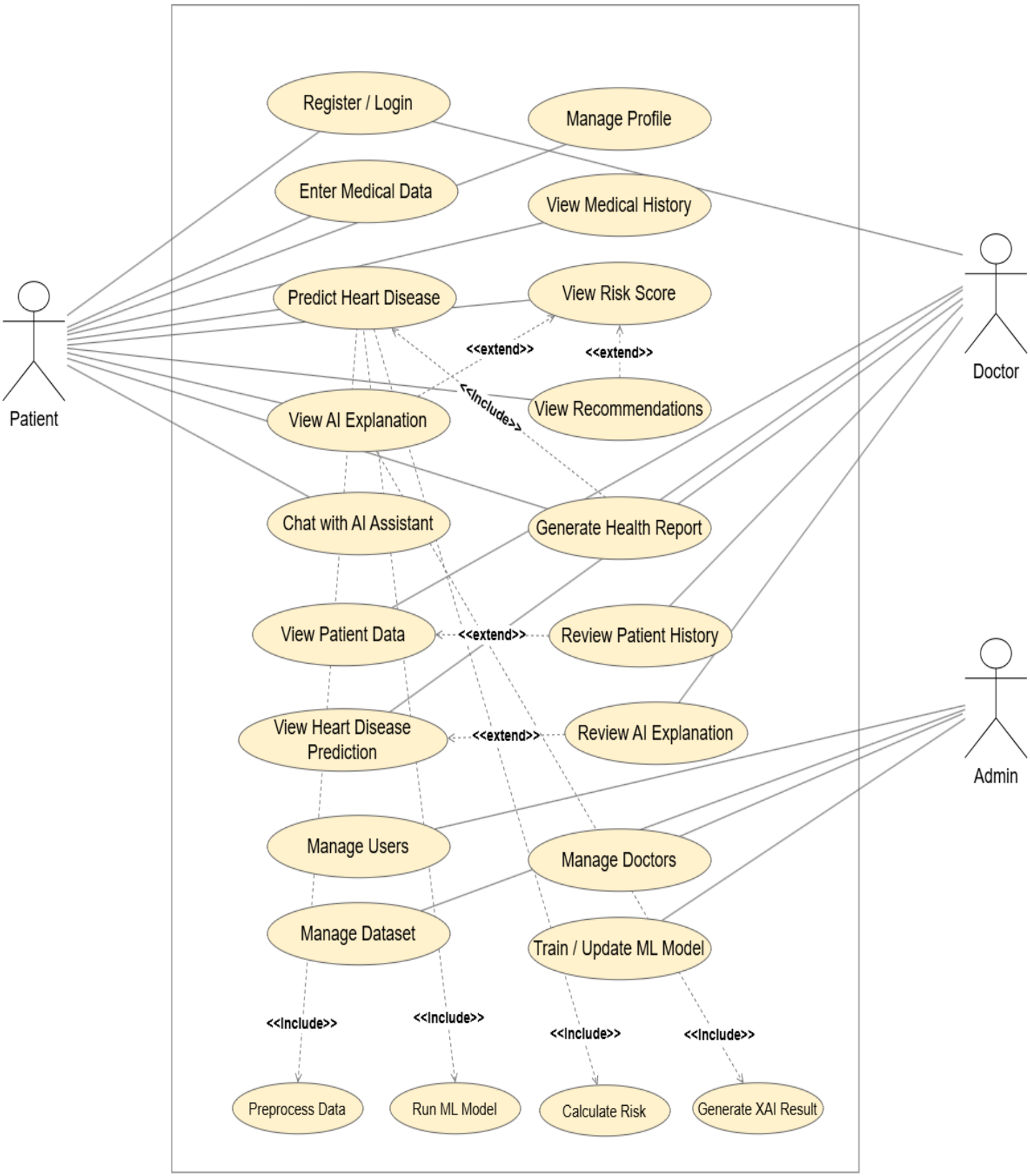
- Processor: Intel Core i3 or equivalent and above
- RAM: Minimum 4 GB; 8 GB or more recommended
- Storage: Sufficient free space for the dataset, patient records, trained machine learning models, and generated PDF reports

1.4.2. Software Requirements

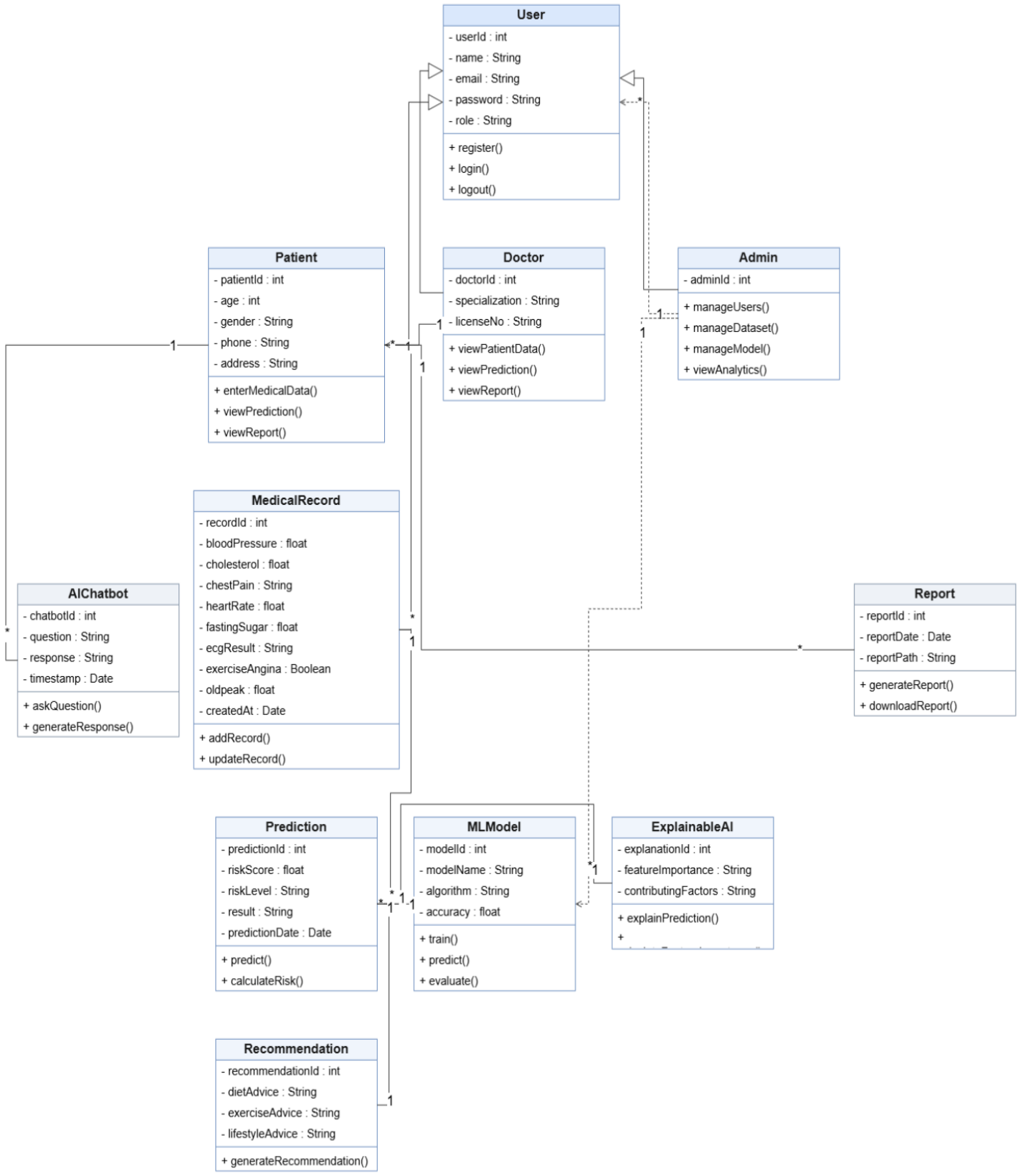
- Operating System: Windows 10/11 or a compatible operating system
- Frontend: React.js, HTML, CSS, and JavaScript
- Backend: Python and Django REST Framework
- Database: SQLite for development; PostgreSQL can be used for deployment
- Machine Learning: Scikit-learn, Pandas, NumPy, and XGBoost
- Explainable AI: SHAP or LIME
- Data Visualization: Recharts or Chart.js
- PDF Generation: ReportLab
- Data Processing: Pandas and NumPy
- Development Tools: Visual Studio Code
- Browser: Google Chrome or Microsoft Edge

2.UML Diagram

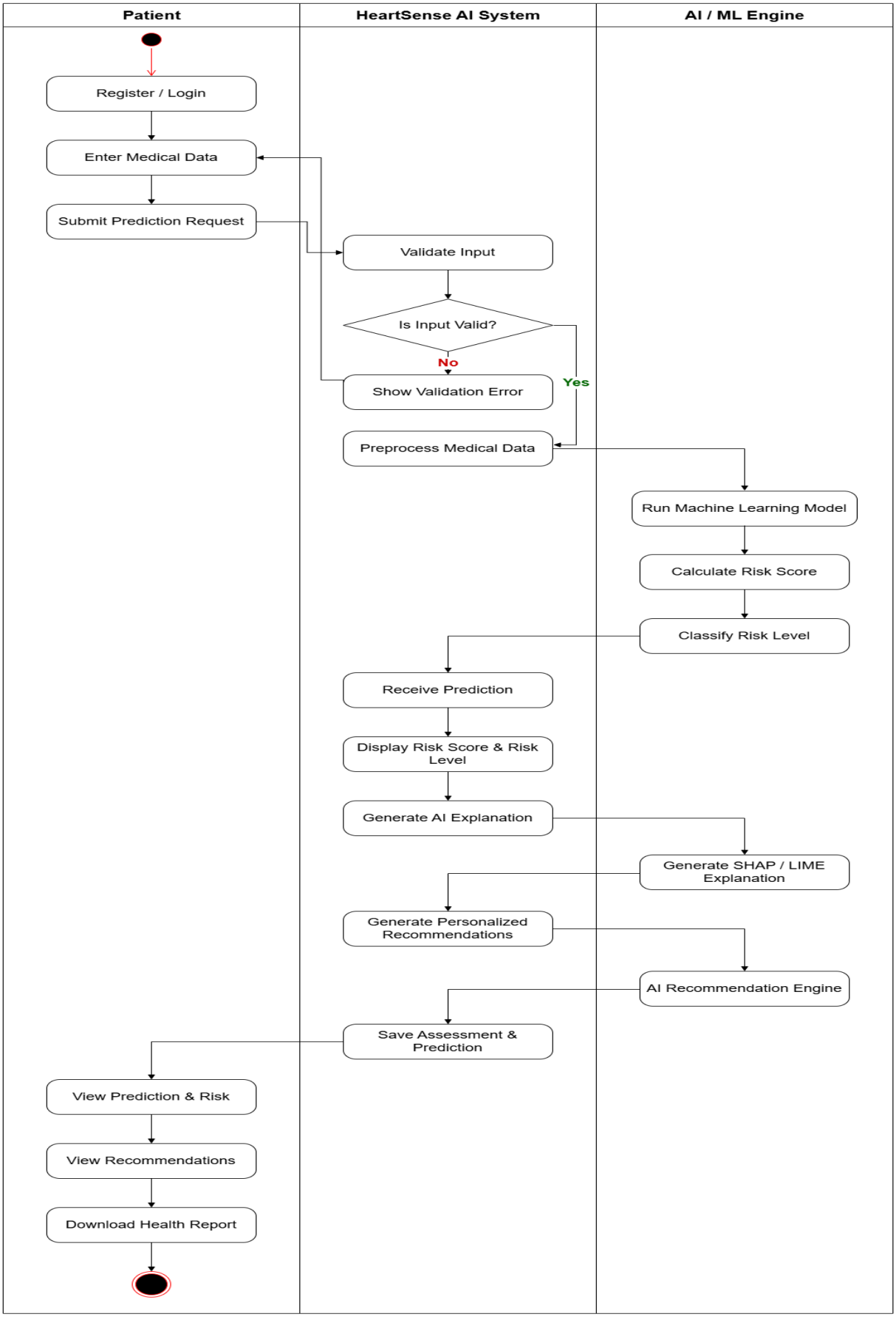
2.1. Use-Case Diagram:



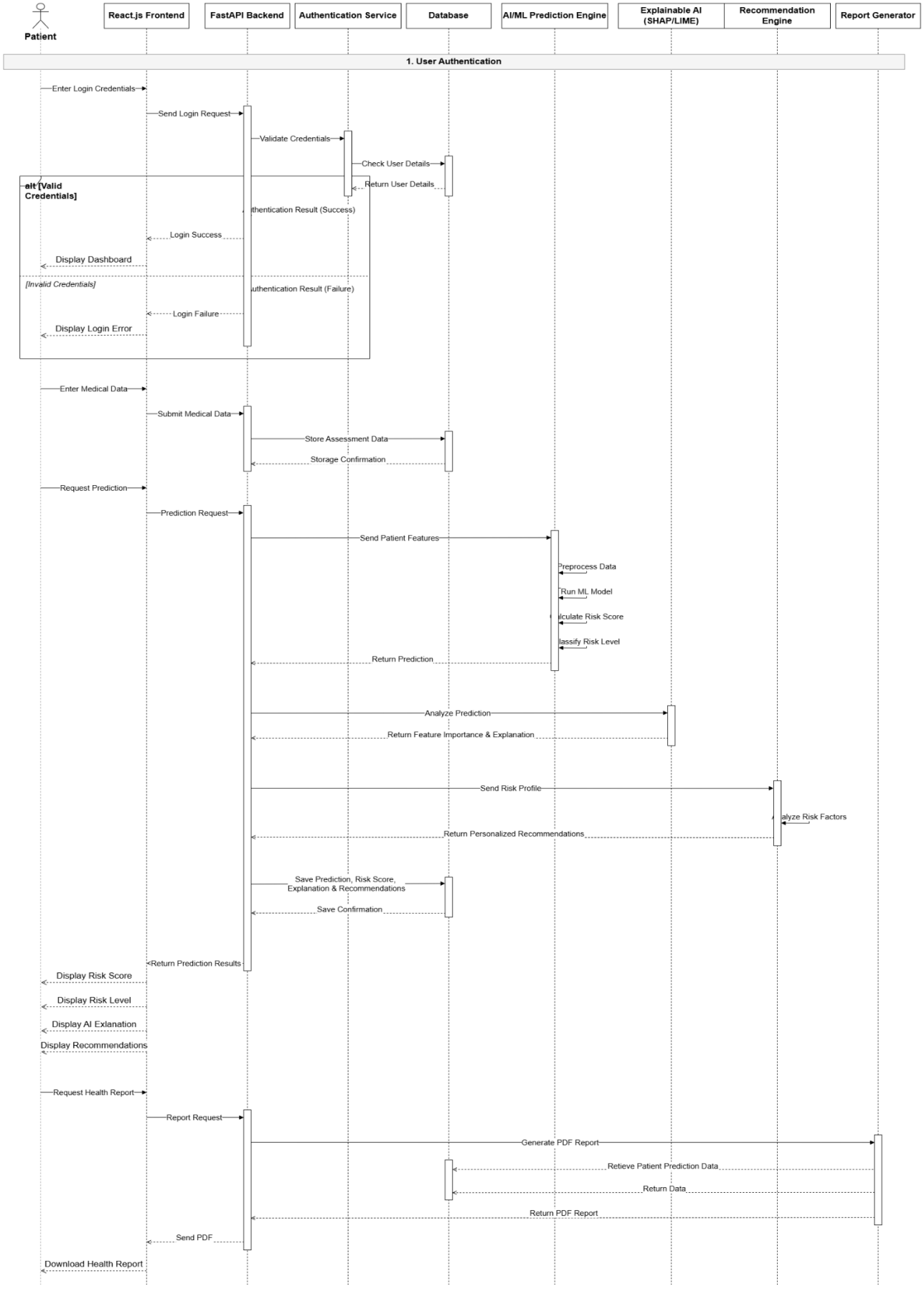
2.2. Class Diagram:



2.3. Activity Diagram:



2.4. Sequence Diagram:



3. Data Dictionary

3.1. User Table

Field Name	Data Type	Key	Description
User_id	INT	PK	Unique user identifier
Full_name	VARCHAR	—	User's full name
Email	VARCHAR	UNIQUE	User email address
Password	VARCHAR	—	Encrypted password hash
Role	ENUM	—	User role: Patient, Doctor, or Admin
Phone	VARCHAR	—	Contact phone number
Created_at	DATETIME	—	Account creation timestamp
Status	BOOLEAN	—	Account status (Active/Inactive)

3.2. Patient Table

Field Name	Data Type	Key	Description
Patient_id	INT	PK	Unique patient identifier
User_id	INT	FK	References User(user_id)
Age	INT	—	Patient age in years
Gender	VARCHAR	—	Gender (Male / Female / Other)
Height	DECIMAL	—	Height in centimeters
Weight	DECIMAL	—	Weight in kilograms
Blood_pressure	VARCHAR	—	Baseline blood pressure reading
Cholesterol	FLOAT	—	Cholesterol level
Fasting_blood_sugar	FLOAT	—	Fasting blood sugar level
Smoking_status	BOOLEAN	—	Smoking history flag
Family_history	BOOLEAN	—	Family history of heart disease
Created_at	DATETIME	—	Record creation timestamp

3.3. Medical Assessment Table

Field Name	Data Type	Key	Description
Assessment_id	INT	PK	Unique assessment identifier
Patient_id	INT	FK	References Patient(patient_id)
Chest_pain_type	VARCHAR	—	Type of chest pain experienced
Resting_bp	FLOAT	—	Resting blood pressure (mmHg)
Cholesterol	FLOAT	—	Serum cholesterol (mg/dl)
Fasting_bs	BOOLEAN	—	Fasting blood sugar > 120 mg/dl
Resting_ecg	VARCHAR	—	Resting ECG results
Max_heart_rate	INT	—	Maximum heart rate achieved
Exercise_angina	BOOLEAN	—	Exercise-induced angina status
Oldpeak	FLOAT	—	ST depression induced by exercise
St_slope	VARCHAR	—	Slope of peak exercise ST segment
Vessels	INT	—	Number of major vessels colored by fluoroscopy
Thalassemia	VARCHAR	—	Thalassemia diagnostic result
Assessment_date	DATETIME	—	Date and time of assessment

3.4. Prediction Table

Field Name	Data Type	Key	Description
Prediction_id	INT	PK	Unique prediction identifier
Assessment_id	INT	FK	References Medical Assessment(assessment_id)
Model_name	VARCHAR	—	ML model used for evaluation
Prediction	VARCHAR	—	Predicted disease outcome
Risk_score	FLOAT	—	Calculated risk percentage (0-100%)
Risk_level	VARCHAR	—	Risk classification (Low, Moderate, High)
Confidence_score	FLOAT	—	Model confidence score
Prediction_date	DATETIME	—	Timestamp of prediction generation

3.5. Explainable AI Table

Field Name	Data Type	Key	Description
Explanation_id	INT	PK	Unique explanation identifier
Prediction_id	INT	FK	References Prediction(prediction_id)
Method	VARCHAR	—	XAI Method used (SHAP / LIME)
Feature_name	VARCHAR	—	Key medical feature evaluated
Feature_value	VARCHAR	—	Patient's feature value
Contribution	FLOAT	—	Feature contribution score to prediction
Explanation_text	TEXT	—	Human-readable explanation text

3.6. Recommendation Table

Field Name	Data Type	Key	Description
Recommendation_id	INT	PK	Unique recommendation identifier
Prediction_id	INT	FK	References Prediction(prediction_id)
Diet_advice	TEXT	—	General dietary recommendations
Exercise_advice	TEXT	—	Physical activity and exercise guidelines
Lifestyle_advice	TEXT	—	General lifestyle modifications
Medical_advice	TEXT	—	Clinical recommendations & specialist care guidance
Created_at	DATETIME	—	Timestamp of recommendation creation

3.7. AI Chat Table

Field Name	Data Type	Key	Description
Chat_id	INT	PK	Unique chat log identifier
User_id	INT	FK	References User(user_id)
Question	TEXT	—	User's query/question text
Ai_response	TEXT	—	AI-generated conversational response
Created_at	DATETIME	—	Chat interaction timestamp

3.8. Report Table

Field Name	Data Type	Key	Description
Report_id	INT	PK	Unique report identifier
Patient_id	INT	FK	References Patient(patient_id)
Prediction_id	INT	FK	References Prediction(prediction_id)
Report_file	VARCHAR	—	PDF report file path or filename
Generated_by	INT	FK	References User(user_id) who generated report
Generated_at	DATETIME	—	Report generation timestamp

3.11. ML Model Table

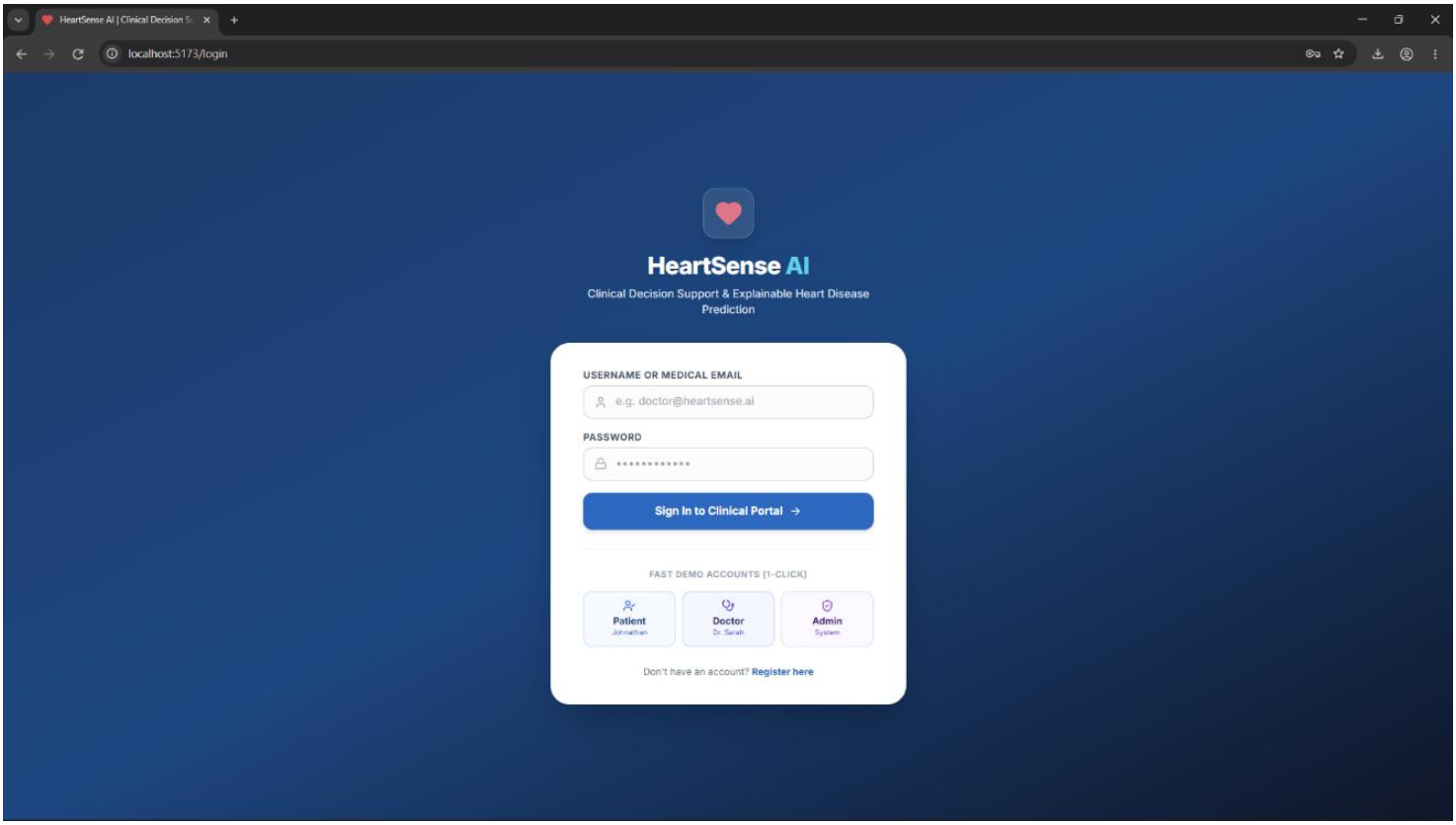
Field Name	Data Type	Key	Description
Model_id	INT	PK	Unique model identifier
Model_name	VARCHAR	—	Model name/identifier
Algorithm	VARCHAR	—	Algorithm (Random Forest, SVM, XGBoost, etc.)
Accuracy	FLOAT	—	Model accuracy score
Precision	FLOAT	—	Model precision score
Recall	FLOAT	—	Model recall score
F1_score	FLOAT	—	F1 score
Auc_score	FLOAT	—	ROC-AUC evaluation score
Version	VARCHAR	—	Model version string
Trained_date	DATETIME	—	Model training completion date
Status	BOOLEAN	—	Model active status flag

3.12. Admin Table

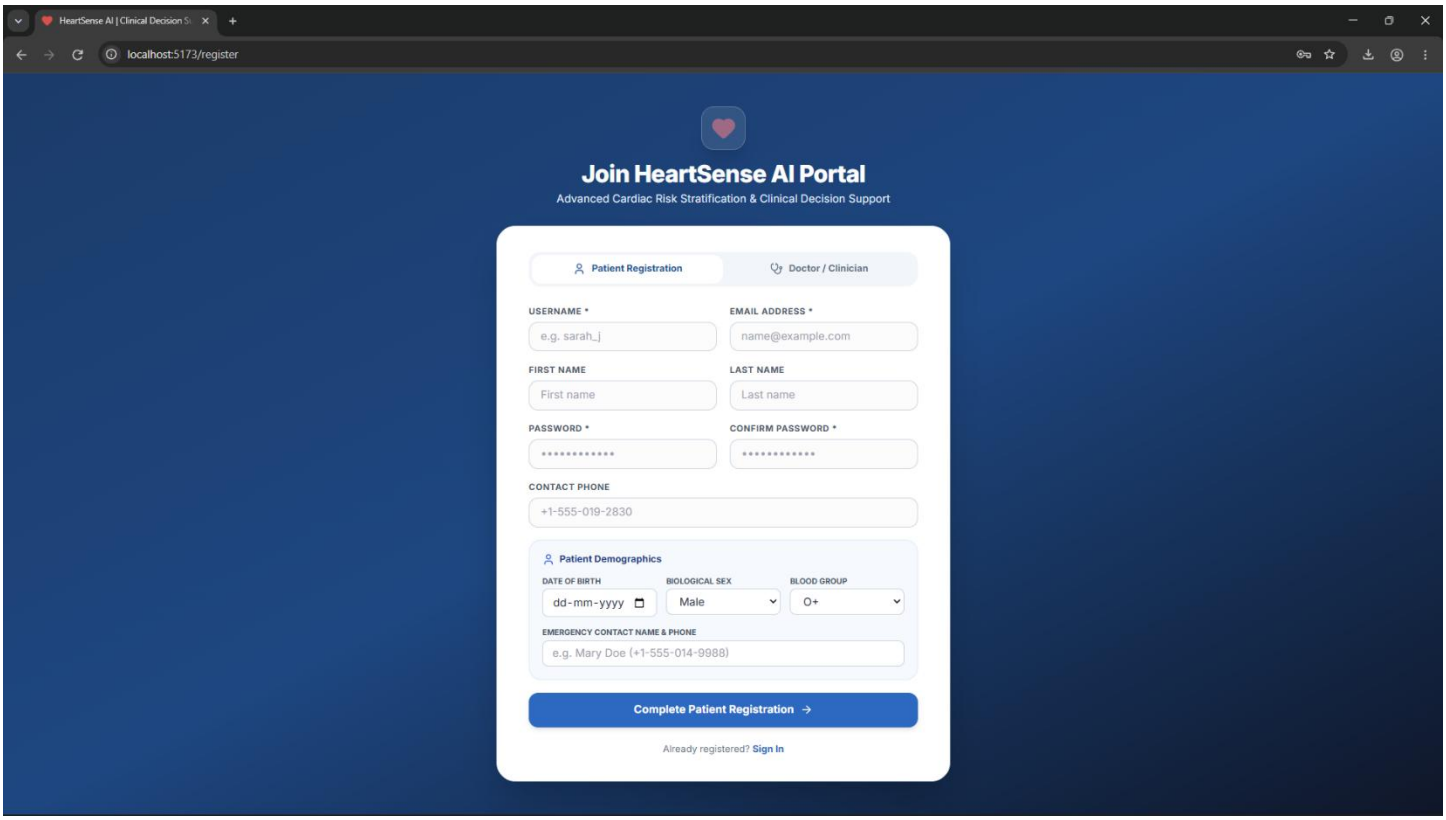
Field Name	Data Type	Key	Description
Admin_id	INT	PK	Unique administrator identifier
User_id	INT	FK	References User(user_id)
Permissions	TEXT	—	JSON/Text specification of permissions
Last_login	DATETIME	—	Last administrative login timestamp
Status	BOOLEAN	—	Admin account active status

4. ScreenShots

4.1. Login Page:

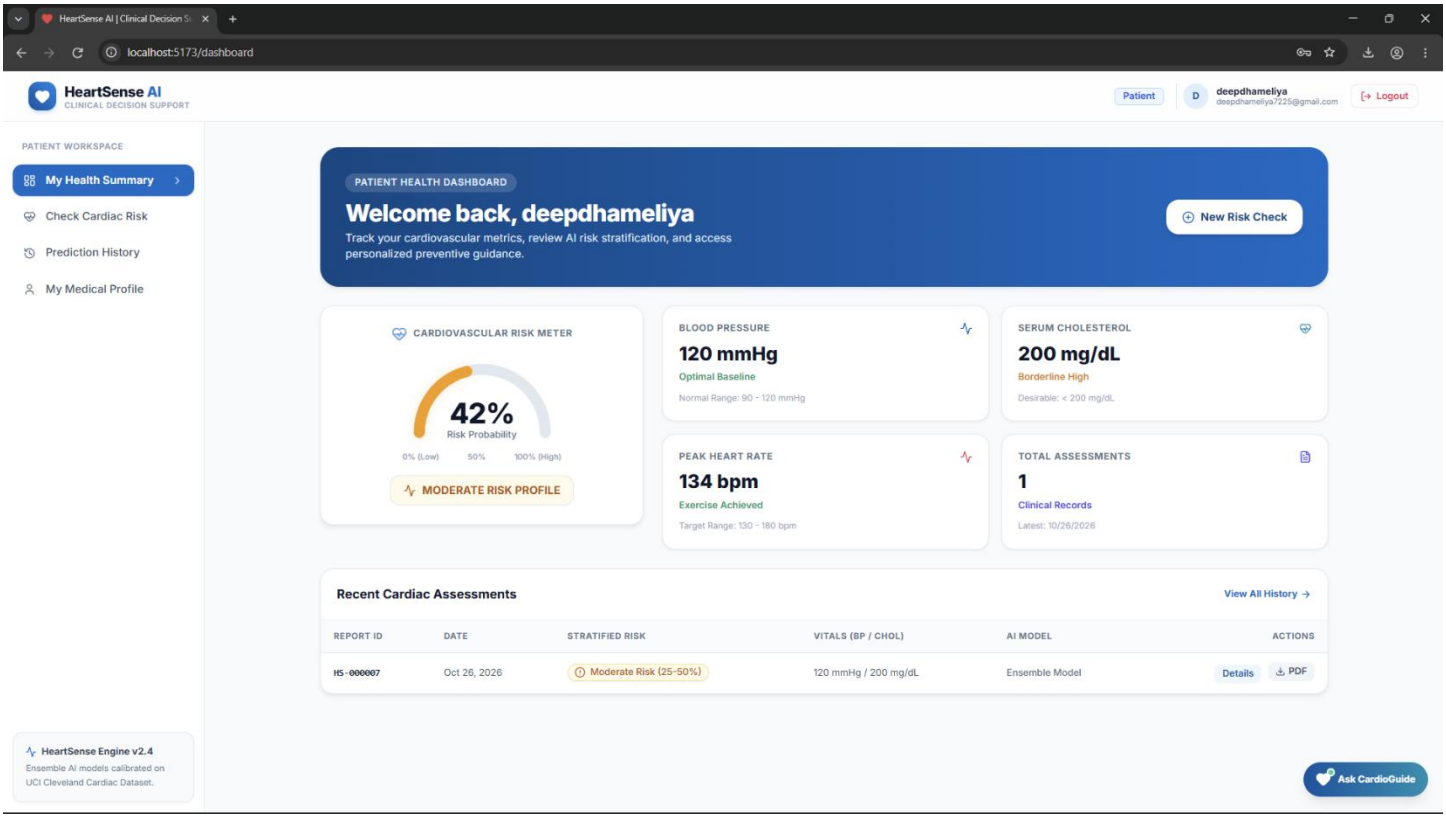


4.2. Register Page:

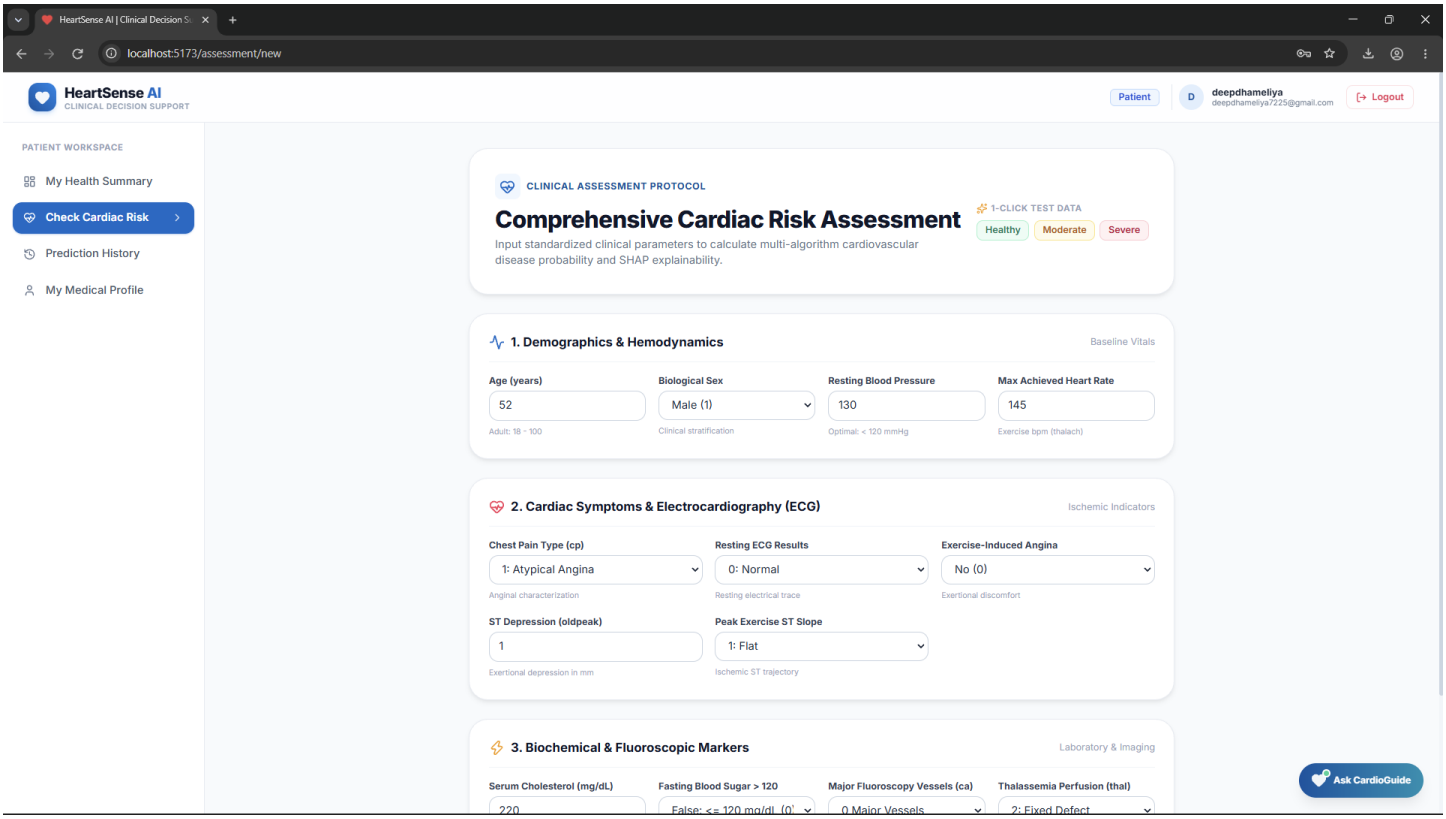


4.3. Patient Side:

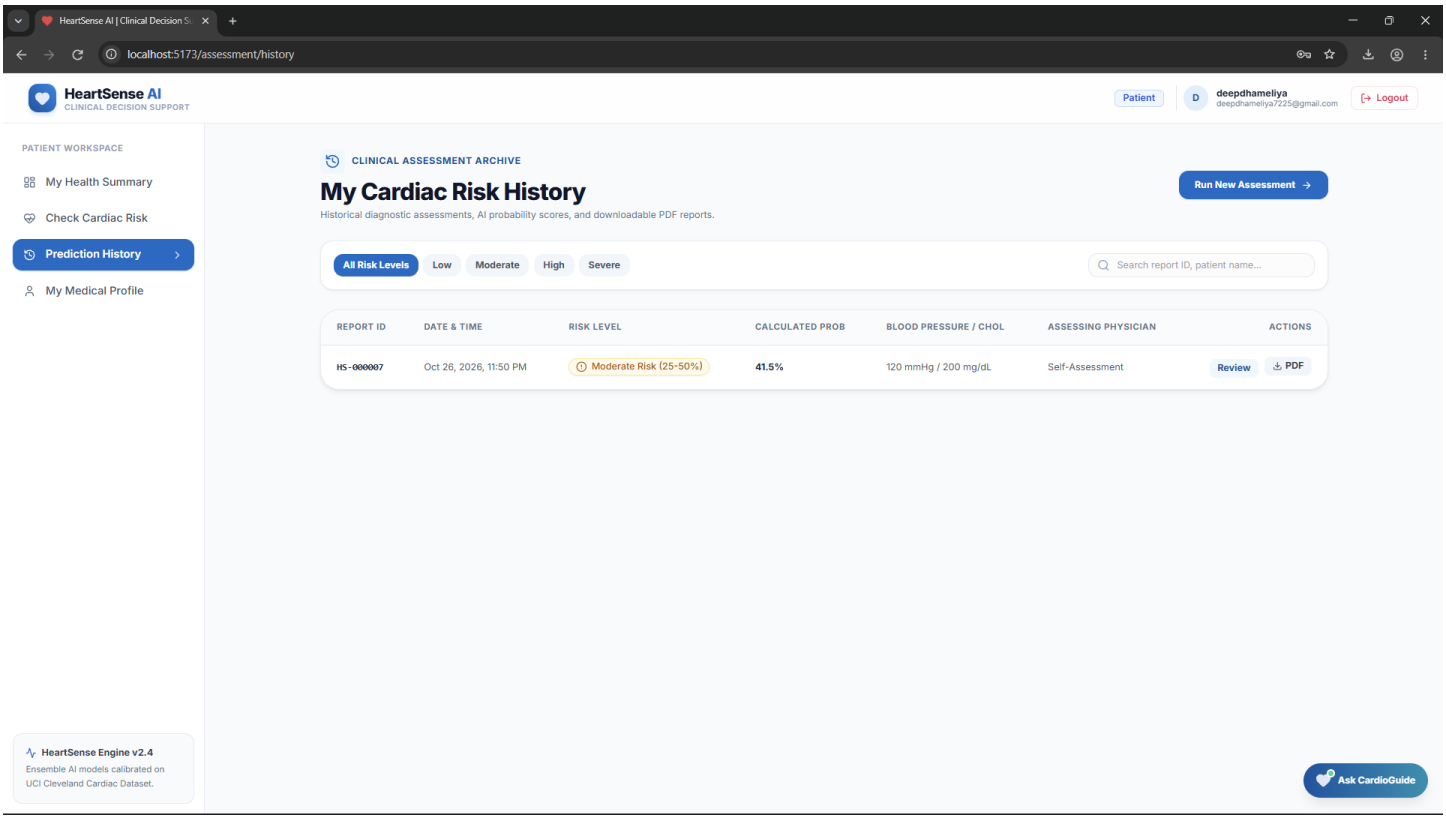
4.3.1. Dashboard Page:



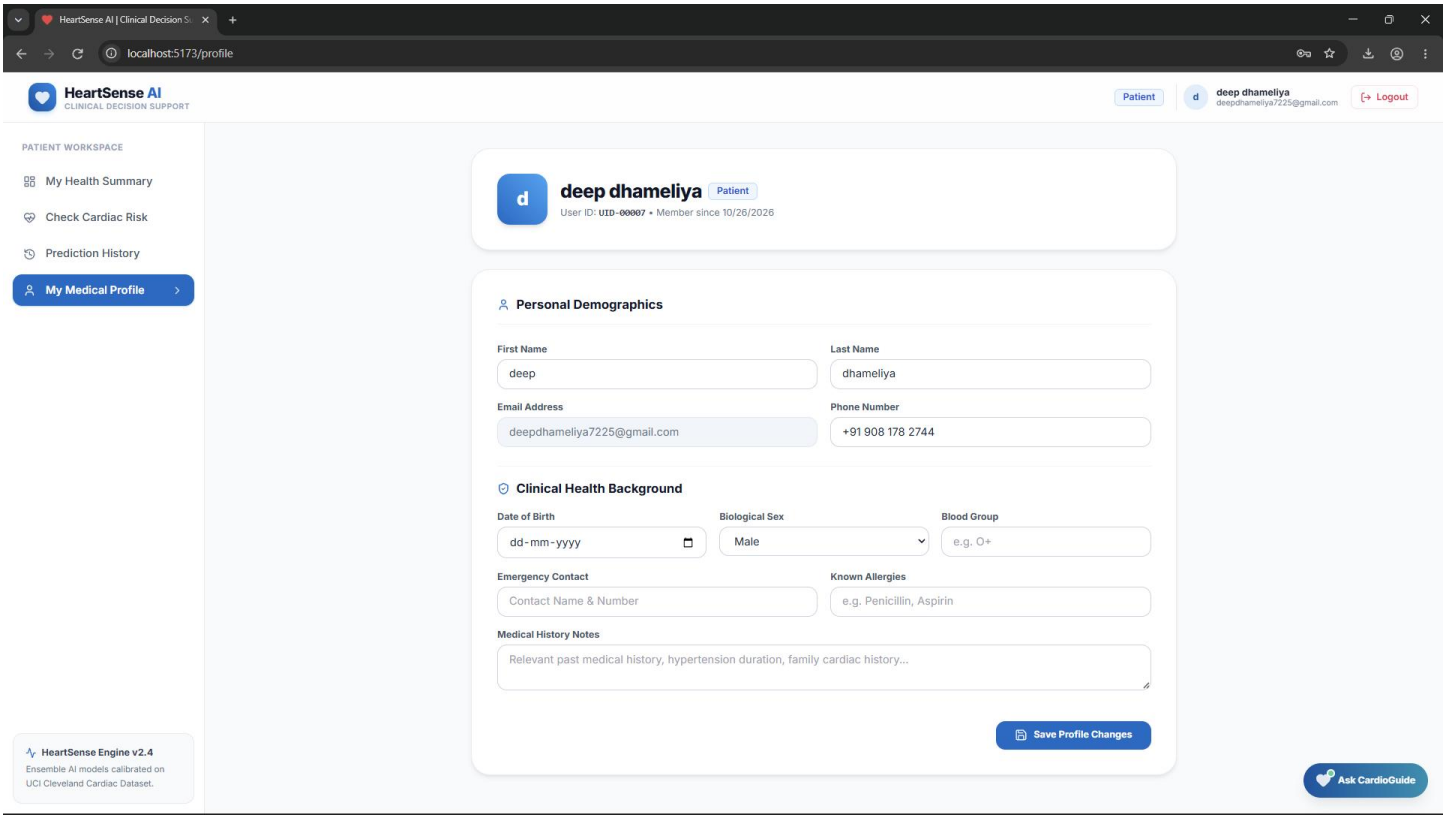
4.3.2. Risk Check Page:



4.3.3. History Page:

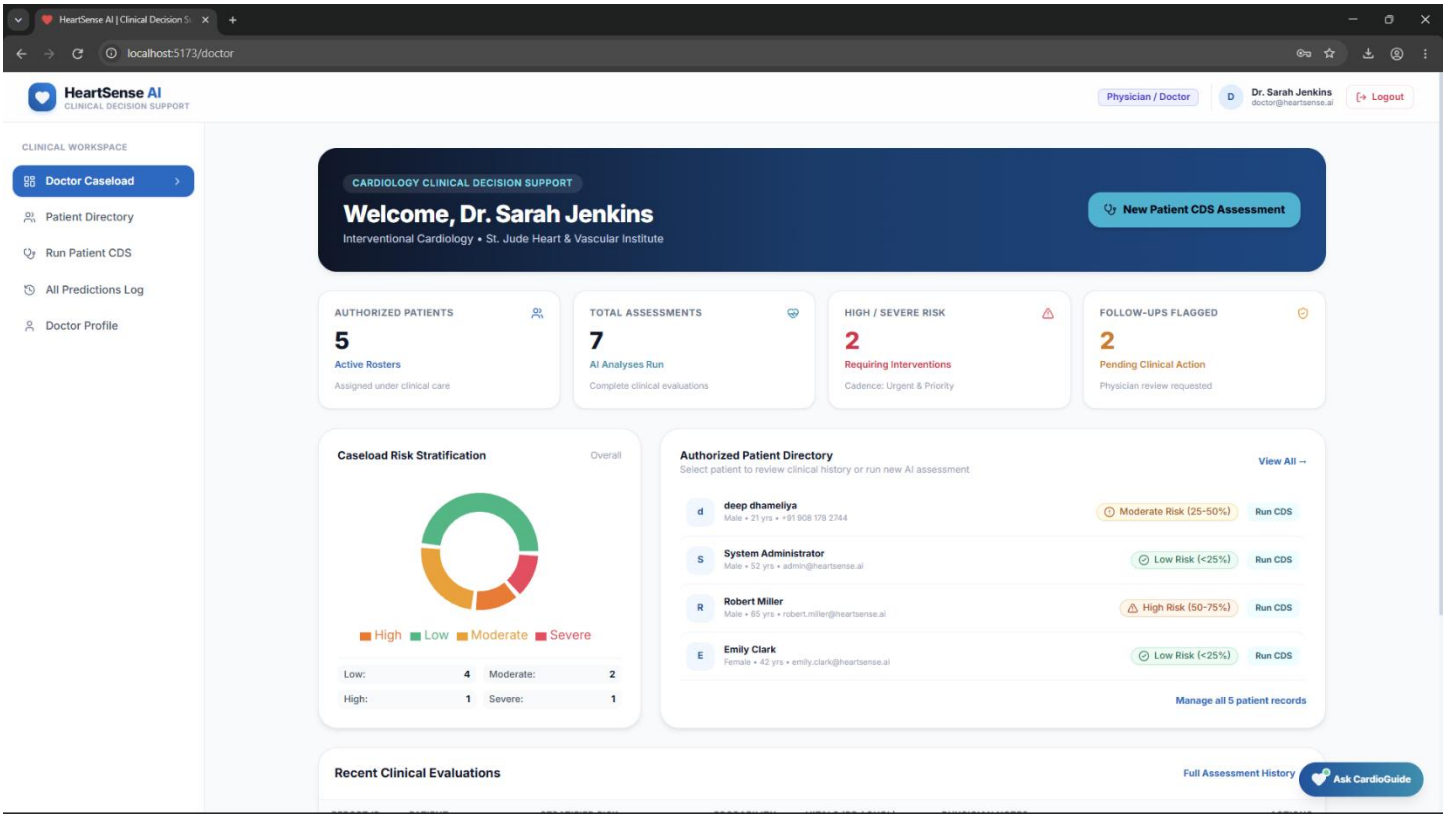


4.3.4. Profile Page:

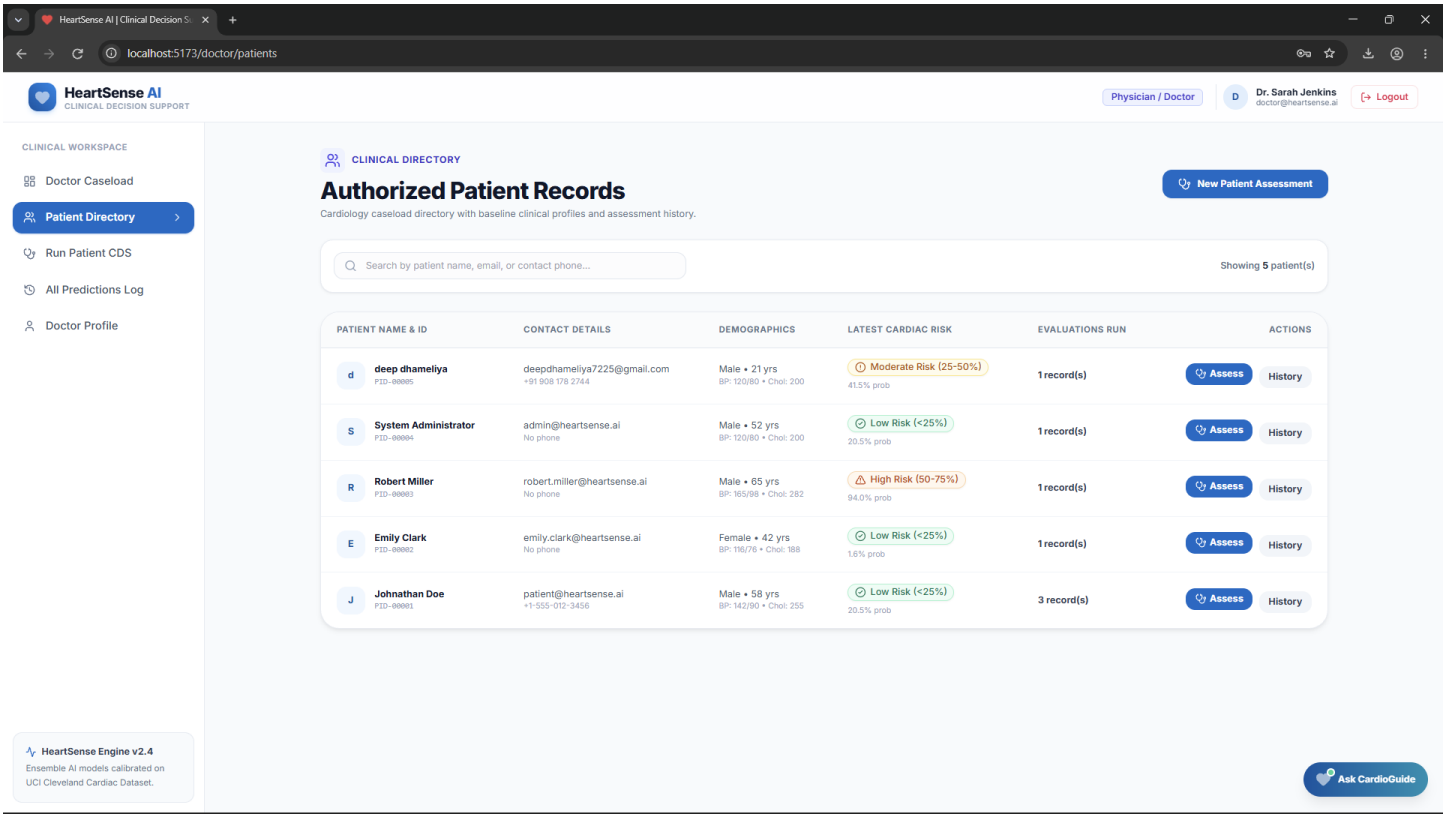


4.4. Doctor Side:

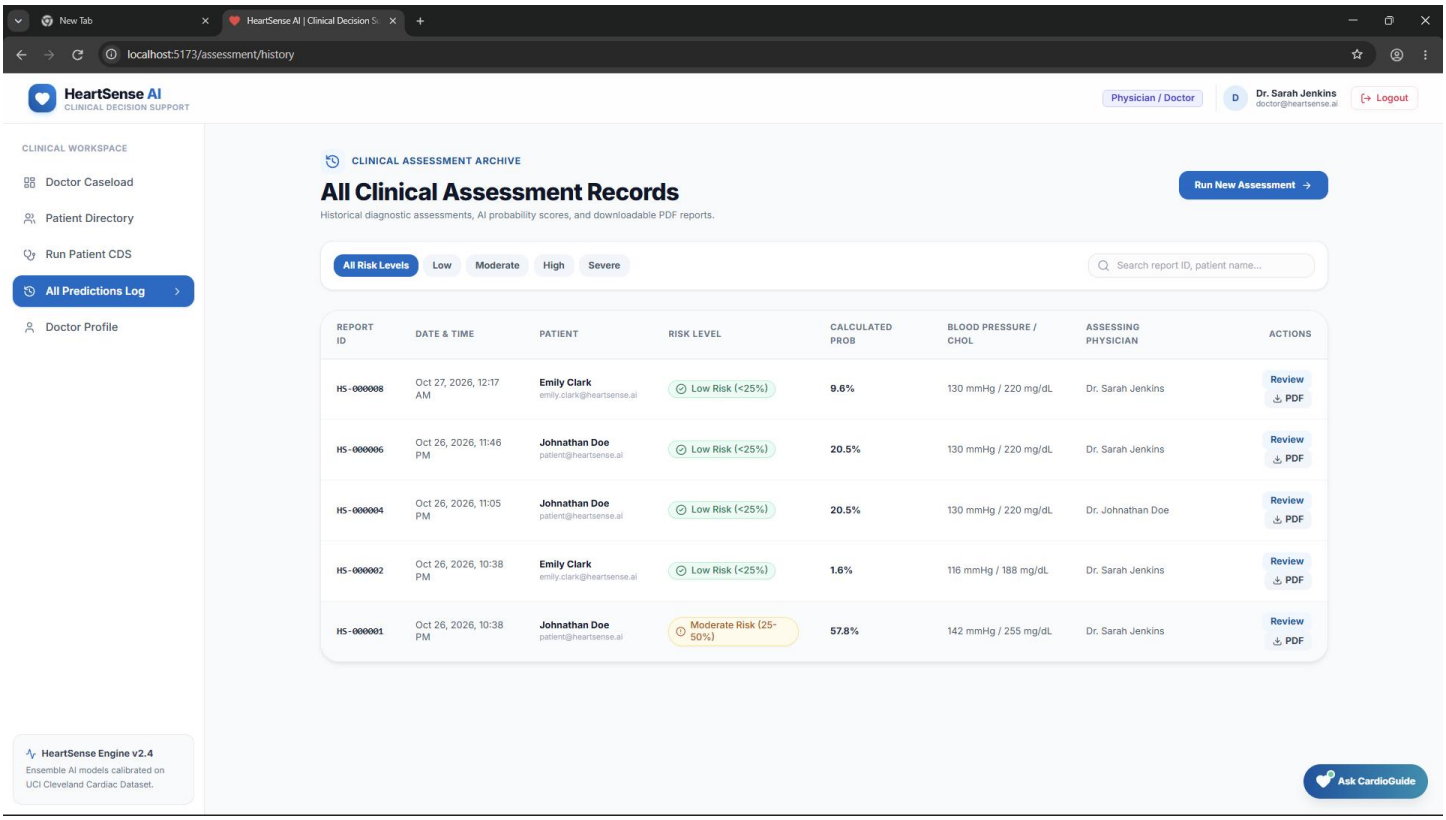
4.4.1. Dashboard Page:



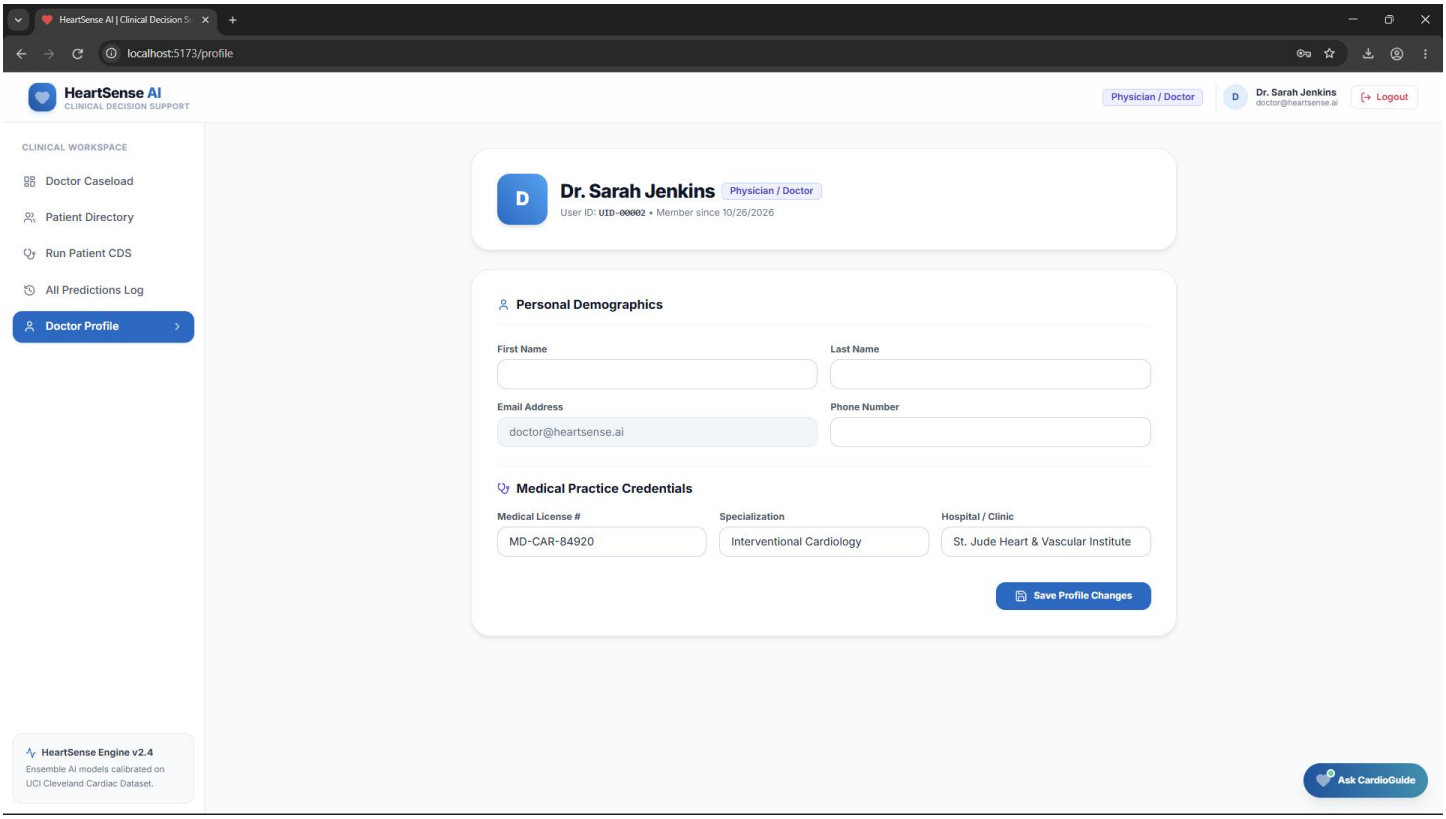
4.3.2. Patients Directory Page:



4.4.3. Clinical Record Page:

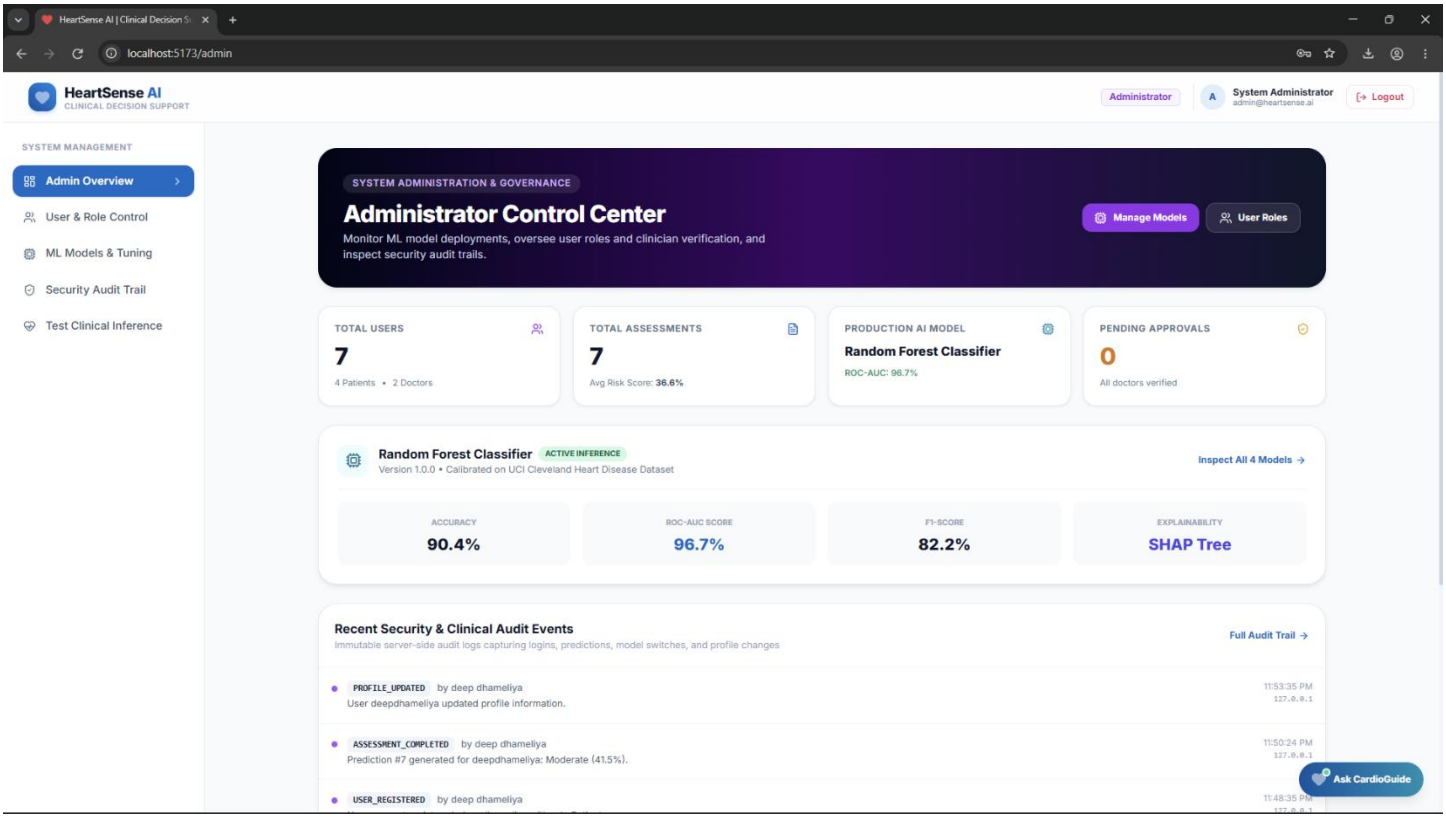


4.4.4. Profile Page:

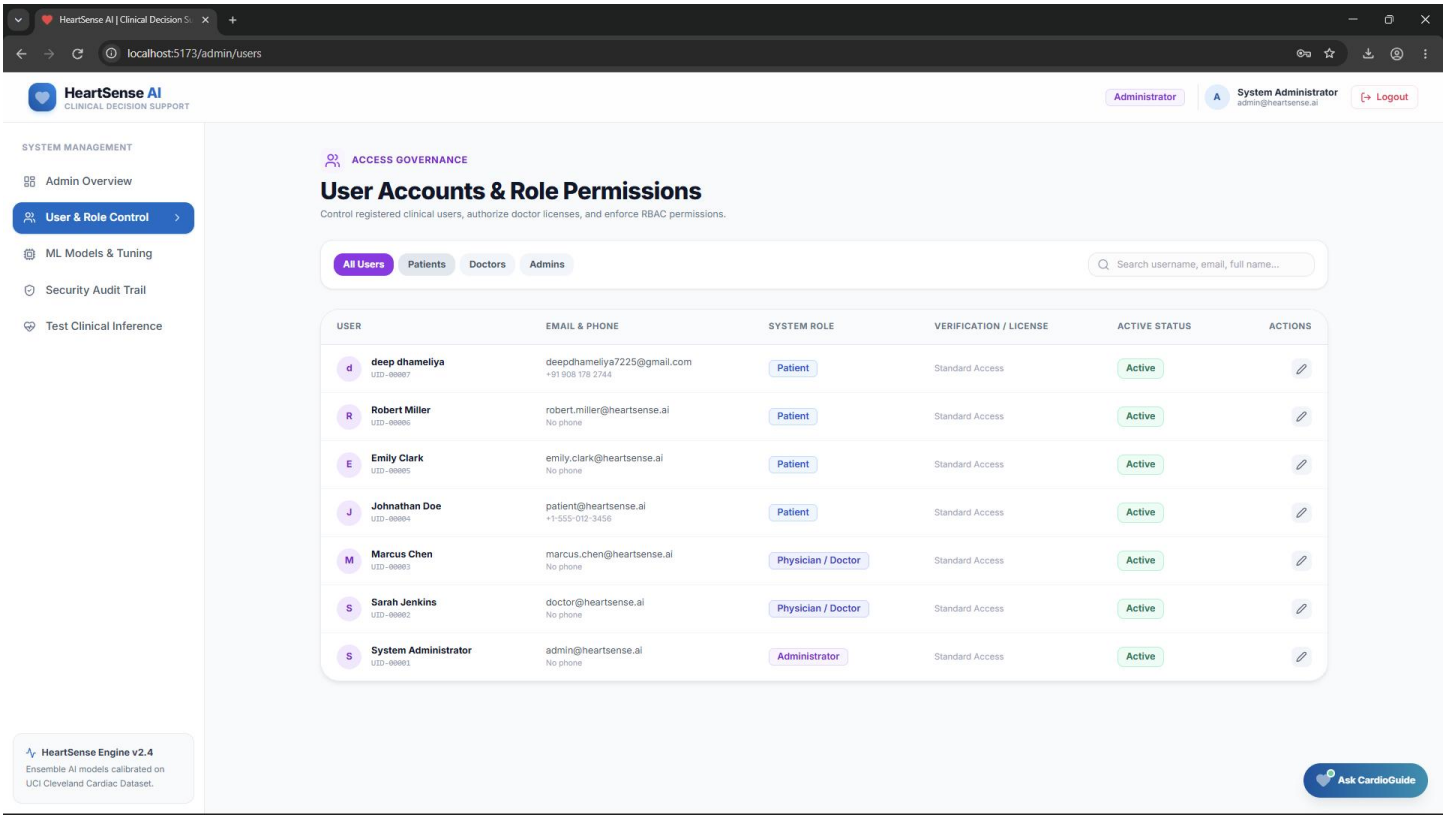


4.5. Admin Side:

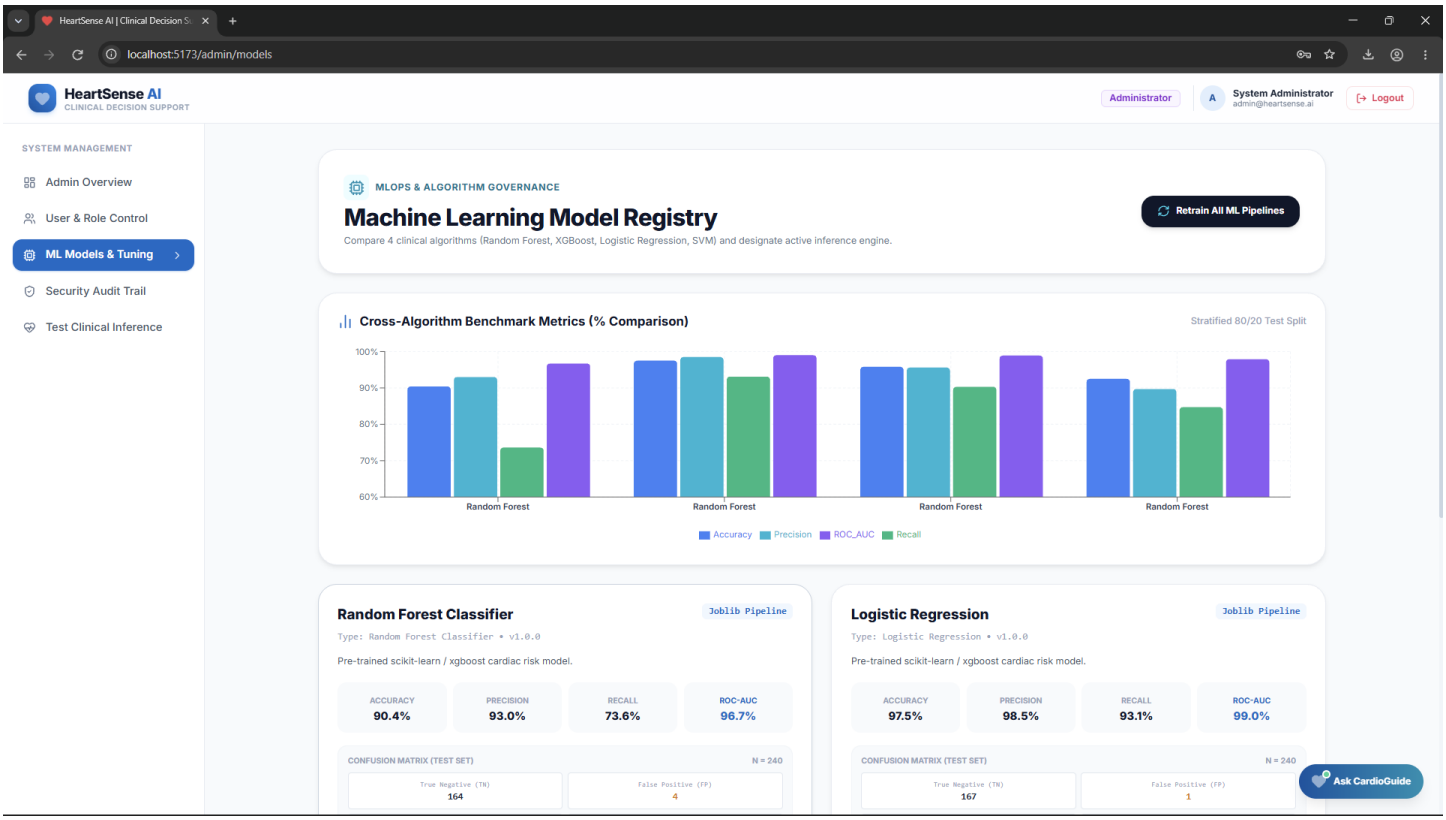
4.5.1. Dashboard Page:



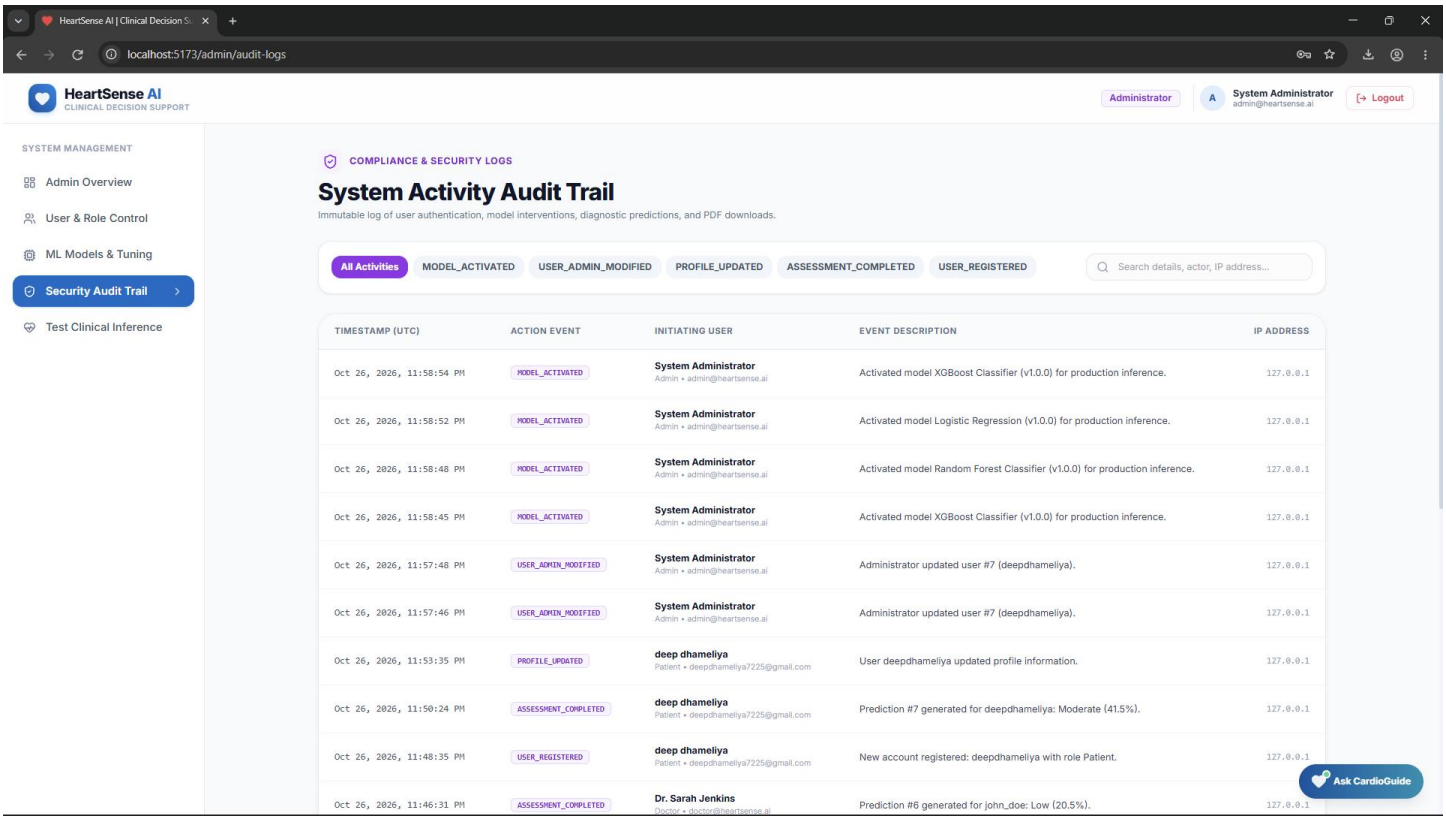
4.5.2. User Management Page:



4.5.3. Model Tuning Page:



4.5.4. Audit Logs Page:



5. References

https://archive.ics.uci.edu/dataset/45/heart+disease
https://docs.djangoproject.com/en/6.1/
https://react.dev/
https://xgboost.readthedocs.io/en/stable/
https://shap.readthedocs.io/en/latest/
https://docs.reportlab.com/
https://docs.python.org/3/